

SELVARAJ M

Aspiring Software Project Engineer Trainee | Java Programmer | DSA & Problem Solving

- Student Ambassador – AICTE IDEA Lab, Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
- Chairperson – IEEE Circuits and Systems (CAS) Society, Vel Tech



Avadi, Chennai, Tamil Nadu, India



selvaisroaslv@gmail.com



8825991678

ABOUT ME

- Electronics and Communication Engineering student (Cybersecurity specialization) with strong programming fundamentals in Java, C, and Python, and a solid grounding in Data Structures & Algorithms, OOP, DBMS, and Operating Systems.
- Consistent record of solving algorithmic and quantitative problems through competitive programming, with hands-on application in dynamic programming, greedy algorithms, and mathematical/logical reasoning.
- Proven ability to translate problem-solving skills into working systems — built and shipped full-stack web applications, REST APIs, and embedded/IoT projects independently and in teams.
- First-prize winner at two international hackathons (VISAI Tech Expo, Pragyan Techfest – NIT Trichy) and Top-37 national finalist at L&T Techgrium Hackathon (64,000+ participants, 540+ institutes), reflecting strong analytical thinking under real-world constraints.

CORE STRENGTHS FOR THIS ROLE

- Programming & Problem-Solving: Java, C, Python — comfortable with algebra-, probability-, and statistics-based problem sets and logical/aptitude reasoning through regular competitive programming practice.
- Core CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks.
- Applied Physics/Systems Thinking: practical exposure to motion, sensing, and real-time system behavior through embedded and robotics projects (see Projects).

EDUCATION

B.Tech – Electronics and Communication Engineering (Specialization: Cybersecurity)

Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology | CGPA: 8.3 (current)

 2023 – Present

Higher Secondary Education (12th Grade), CBSE

Ramco Vidya Mandir Senior Secondary School — 70%

 2021 – 2023

Secondary School Education (10th Grade), CBSE

Vidya Mandir Senior Secondary School — 88%

SKILLS

- **Technical Skills**
- **Programming Languages:** Java, C, Python, JavaScript, SQL
- **Core CS Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks
- **Java Full Stack:** Core Java, JDBC, REST APIs, Spring Boot (Basics), Spring MVC (Basics)
- **Web Development:** MongoDB, Express.js, React.js, Node.js (MERN), HTML5, CSS3, Bootstrap
- **AI/ML & Computer Vision:** OpenCV, TensorFlow, Scikit-learn, Real-time Object Detection, Image Processing
- **Tools & Platforms:** Git, GitHub, REST APIs, Postman, Linux, VS Code
- **Cybersecurity & Embedded:** Network Security, Basic Cryptography, Secure Coding Practices, ESP32, Raspberry Pi, Sensor Integration

PROJECTS

Full Stack Web Application Development

- Developed responsive web applications using HTML5, CSS3, JavaScript, React.js, Node.js, and Express.js.
- Designed RESTful APIs and integrated frontend with MongoDB/SQL databases for CRUD operations.

- Implemented user authentication, responsive UI, and optimized application performance.

ESP32-Based Smart Health Monitoring System

- Developed an ESP32-based embedded health monitoring system using Embedded C with MAX30102, AD8232 ECG, and DS18B20 sensors.
- Implemented I2C, ADC, and UART communication for real-time physiological data acquisition and processing.
- Integrated ThingSpeak cloud platform for remote monitoring and developed a multilingual (10-language) web dashboard with downloadable reports.

AI-Based Agri Bot System

- Developed an IoT-enabled smart farming system using ESP32 with real-time sensor data acquisition and automated irrigation monitoring.
- Integrated environmental sensors and embedded firmware for autonomous monitoring and decision-making.
- Applied computer vision techniques for crop and weed detection to support precision agriculture.
- Enabled remote monitoring through IoT-based data transmission and visualization.

Smart Retail Monitoring System

- Developed a computer vision-based inventory monitoring system for automated product detection and stock tracking.
- Implemented object detection algorithms for real-time shelf monitoring and inventory management.
- Generated automated reports and analytics to improve inventory accuracy and operational efficiency.

INTERNSHIP EXPERIENCE

Junior Full Stack Developer & System Analyst Intern — Universiti Sains Islam Malaysia (USIM), Malaysia

- Worked on frontend and backend web application development; developed and integrated REST APIs.
- Assisted in system analysis, workflow optimization, and application enhancement; worked with databases and debugging.

Project Intern – Agri Bot (AI-Based Smart Farming System) — Ashok Leyland, India | Mar 2024 – Jun 2024

- Developed an AI-based smart farming rover using embedded systems and IoT technologies.
- Applied problem-solving and analytical skills for system optimization and real-time data handling.

KEY ACHIEVEMENTS

- 🏆 **First Prize – Pragyan International Techfest, NIT Trichy**
- 🏆 **First Prize – VISAI International Hackathon, Vel Tech University**
- 🏆 **Special Prize – TAFE Perambur Plant (Innovista '25)**
- 🏆 **Finalist – L&T Techgium Hackathon (2026)**
 - Selected among Top 37 finalist teams nationwide
 - Competed among 64,000+ students from 540+ institutes across India